Week 9: Analysis of Results

Empirical Research Methods

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Learning Objectives

- Understand the difference between Intention-to-Treat and Treatment-on-Treated analysis
- Learn how to estimate treatment effects from experimental data
- Understand heterogeneous treatment effects
- Learn to conduct robustness checks and sensitivity analysis
- Interpret results in context of your pre-analysis plan

From Data to Results

1. Clean and prepare data
2. Check balance between treatment and control and check compliance
3. Estimate treatment effects using pre-specified methods
4. Test for heterogeneous effects
5. Conduct robustness checks
6. Interpret results and discuss limitations

Follow your pre-analysis plan. Don't change methods based on results (HARKing, p-hacking, etc).

Analysis Tools: Overview of the Basics

Duru and Kopper 2023

Average Treatment Effect (ATE)

The average causal effect of the treatment on the population, as measured via the outcome variable.

Local ATE (LATE)

The average treatment effect *only* for those that follow their assigned treatment.

Intention to Treat (ITT)

The average treatment effect on how the treatment were originally assigned, *not* what the units actually got.

Treatment-on-Treated (ToT)

The treatment effect on those that actually took up the treatment

Classifying Units

Duru and Kopper 2023

A Reality of Experiments

Assigning a unit to a treatment group does not mean it will undergo that treatment. There are four ways to categorized a unit.

- **Compliers**: Unit actually undergoes the procedure assigned to it in the design phase.
- **Always-Takers**: A unit which always takes a treatment, even if in the control group.
- **Never-Takers**: A unit which will never undergo a treatment (i.e. always "wants" to be a control group unit).
- **Defiers**: Do the opposite of whatever the assignment was in the design.

Key Upshot: We *must* assume defiers to not exist, or else we would never be able to calculate an effect, as defiers and compliers could cancel each other out.

Average Treatment Effect

Heinze-Deml 2021, Barrett, D'Agostino McGowan, and Gerke 2026

The ATE estimator can be considered as the examination of the counterfactual by comparing treatment and control.

The ATE Estimator:

$$\text{ATE} = E[Y(1)] - E[Y(0)]$$

where $Y(n)$ represents the treatment status that actually occurred.

Without perfect compliance, this method becomes biased. This can be accounted for in regression models, but not fully resolved.

Local Average Treatment Effect

Duru and Kopper 2023

When dealing with non-compliance, a straightforward choice can be to exclude those that did not comply. This gives us the Local ATE.

The LATE Estimator

$$\text{LATE} = \frac{E[Y|Z=1] - E[Y|Z=0]}{E[d|Z=1] - E[d|Z=0]}$$

- It can only uncover the effects for the compliers.
- Reliers on additional assumptions:
  - Being assigned to treatment does not make you less likely to be treated
  - Unites respond to treatment, not assignment

Intention-to-Treat (ITT) Analysis

Since LATE only works for the outcome, not the assignment, it can be considered too narrow. Therefore, we look at our randomization instead

The ITT Estimator:

$$\text{ITT} = E[Y|Z = 1] - E[Y|Z = 0]$$

where Z is the random assignment (not actual treatment).

While the ITT relies on the same assumptions as LATE, it also:

- Preserves randomization
- Unbiased even with non-compliance
- **Reflects real-world policy impact (some won't comply)**

Treatment-on-Treated (ToT) Analysis

To examine how the treated units changed their behaviour, we can look at the effect of the treatment on only those labs *if there are no labs in the control group that took the treatment*.

The ToT Estimator:

$$\text{ToT} = \frac{\text{ITT}}{\text{Compliance Rate}}$$

What does the ToT tell us?

- Secondary analysis (ITT is primary)
- Requires Instrumental Variables (IV) to be valid
- Estimates effect on compliers, not population average *when non-compliance is not related to potential outcomes*

Estimating Treatment Effects

The Simplest Approach

Compare mean outcomes between treatment and control groups:

$$\hat{\tau} = \bar{Y}_{\text{Treatment}} - \bar{Y}_{\text{Control}}$$

Example: Energy Savings Program

	Treatment	**Control**
Mean Energy Use (kWh)	450	500
Standard Deviation	80	85
Sample Size	250	250

Result: Treatment effect = 450 - 500 = -50 kWh (10% reduction)

Regression Estimation

Simple Regression

$$Y_i = \alpha + \tau \cdot T_i + \epsilon_i$$

where T_i is treatment indicator.

Regression with Covariates

$$Y_i = \alpha + \tau \cdot T_i + \beta X_i + \epsilon_i$$

where X_i are baseline characteristics.

Advantages:

- Reduces variance by controlling for predictive variables
- Improves precision of treatment effect estimate
- Allows for heterogeneous effects

Heterogeneous Treatment Effects

What to look for:
Treatment effects may vary across subgroups. The average effect may hide important heterogeneity.

Interaction Regression: δ is the interaction coefficient

$$Y_i = \alpha + \tau \cdot T_i + \gamma \cdot X_i + \delta \cdot T_i \times X_i + \epsilon_i$$

Example: Does energy program work equally for all income levels?

- Low income: -30 kWh
- Middle income: -50 kWh
- High income: -70 kWh

Statistical Significance

P-value

Probability of observing effect this large (or larger) if null hypothesis is true. A typical threshold is 0.05.

- $p < 0.05$: Reject null, effect is statistically significant
- $p \geq 0.05$: Fail to reject null, insufficient evidence

Confidence Intervals

Range of plausible values for true effect.

$$95\%\ \text{CI} = \hat{\tau} \pm 1.96 \times \text{SE}(\hat{\tau})$$

Interpretation: If we repeated the study N times, the true effect would fall in this range 95% of the time. Stats with Brian 2025

Balance and Robustness Checks

Barrett, D'Agostino McGowan, and Gerke 2026

Why perform these checks?

Ensure results don't depend on arbitrary choices (sample, specification, outliers).

Balance Check

Compare baseline characteristics between treatment and control groups. This can be done via a visual inspection or a test.

Common Robustness Checks:

- **Specification:** Different control variables, functional forms
- **Outliers:** Exclude (justifiably) extreme values, use robust regression
- **Subsamples:** Analyze different time periods or populations
- **Alternative outcomes:** Use related but different outcome measures
- **Clustering:** Adjust standard errors for clustering

Multiple Hypothesis Testing

Bishop and Thompson 2023

The Problem with Multiple Testing

Testing many hypotheses increases the probability of false positives. If you test 20 hypotheses at 5% level, expect 1 false positive by chance.

Worst Case Scenario: independent analyses

$$
\begin{aligned}
P(\text{Type I error in } m \text{ questions}) &= 1 - P(\text{No Type I error in all } m \text{ questions}) \\
\textit{assuming independence} \rightarrow &= 1 - P(\text{No Type I error in } Q_i)^m \\
&= 1 - (1-\alpha)^m
\end{aligned}
$$

What to control

Family-Wise Error Rate	False-Discovery Rate
Consequence of false positive high	Consequence of false positive low
Bonferroni, Bonferroni-Holm	Benjamini-Hochberg, B–Yekutieli

Interpreting Results

Practical vs statistical significance: Consider how what your results imply. Use the parameters of a statistically significant model to forecast something, and see if it means anything.

What to Report

- Point estimate, confidence interval, error estimates
- Whether study was powered to detect realistic effects
- Discuss practical significance, not just statistical significance

For statistically insignificant results, no effect found $\neq$ **No effect exists**

Possible Explanations:

- True effect is zero
- Study is underpowered (MDE too large, too small sample, etc)
- Implementation failed (treatment not delivered)
- Measurement error is too large

Analysis in Your Pre-Analysis Plan

What Should Be Specified

Your PAP Should Include:

- **Primary outcome:** Which outcome is most important?
- **Estimation method(s):** ITT, ToT, regression?
- **Control variables:** Which baseline characteristics to control for?
- **Subgroup analysis:** Will you examine heterogeneous effects?
- **Robustness checks:** What alternative specifications will you test?
- **Multiple testing:** How will you handle multiple hypotheses?

Summary: Analysis of Results

Key Concepts

- Consider the realities of assignment and treatment uptake when planning your analysis
- Regression with covariates improves precision
- Heterogeneous effects reveal whether treatment works for everyone
- Robustness checks ensure results are not artifacts of arbitrary choices

Next Steps

- Clean and prepare your data
- Check balance between groups
- Estimate treatment effects using pre-specified methods
- Conduct robustness checks
- Interpret results in context of your research question

Citations and Resources I

- Barrett, Malcolm, Lucy D'Agostino McGowan, and Travis Gerke (Mar. 2026). *Causal Inference in R*. URL: https://www.r-causal.org/.
- Bishop, Dorothy V. M. and Paul A. Thompson (Mar. 2023). *Evaluating What Works*. URL: https://bookdown.org/dorothy_bishop/Evaluating_What_Works/ (visited on 03/23/2026).
- Duru, Maya and Sarah Kopper (Aug. 2023). *Data analysis*. en. URL: https://www.povertyactionlab.org/resource/data-analysis (visited on 04/24/2026).
- Heinze-Deml, Christina (Apr. 2021). *08_potential_outcomes_counterfactuals_2804.pdf*. en. URL: https://polybox.ethz.ch/index.php/s/7IFu694Hvi4Lxvi (visited on 04/24/2026).

Citations and Resources II



Stats with Brian (Mar. 2025). *I'm 95% Confident You Don't Know What a Confidence Interval Means*. URL: https://www.youtube.com/watch?v=JyByW8YURGU (visited on 04/24/2026).